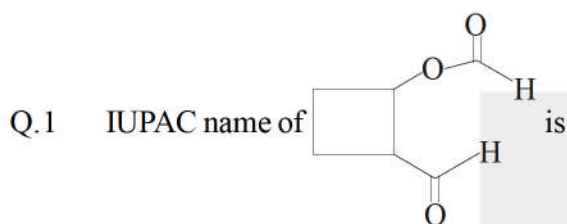


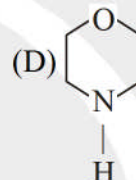
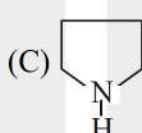
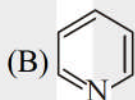
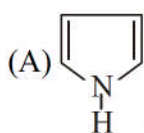
WORK SHEET - 02

Single correct :

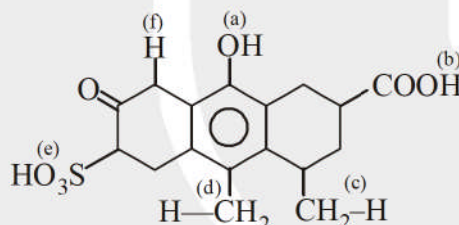


- (A) Cyclobutane-1-carbaldehyde-2-carboxylate
(B) 2-Formyl cyclobutane carboxylate
(C) 2-Formylcyclobutyl methanoate
(D) 2-Methanoyloxy cyclobutane carbaldehyde

Q.2 Which of the following will have least pK_b value.

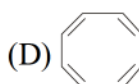
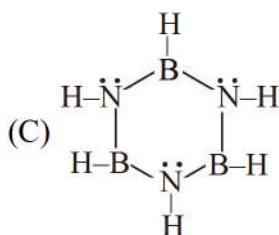
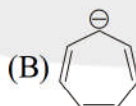
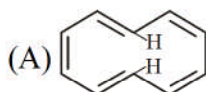


Q.3 Arrange the following in increasing order of acid strength.



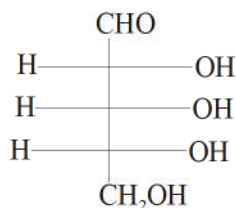
- (A) $c < f < d < a < b < e$
(B) $f < d < c < b < e < a$
(C) $f < c < d < b < e < a$
(D) $c < d < f < a < b < e$

Q.4 Which of the following is aromatic?

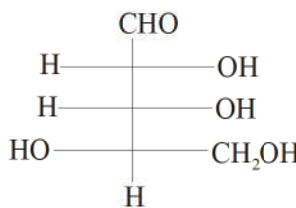


Comprehension :

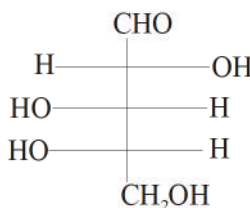
Paragraph for question nos. 5 to 7



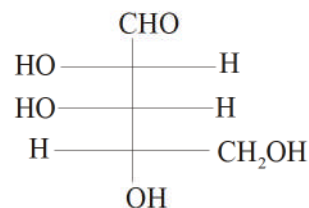
(I)



(II)



(III)



(IV)

Q.5 Which of the above compounds are enantiomers?

- (A) I and III (B) II and III (C) II and IV (D) I and IV

Q.6 Which of the above compounds are identical?

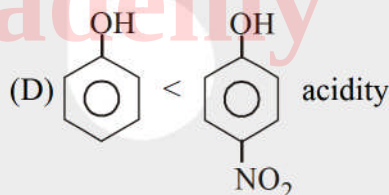
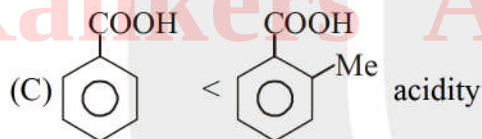
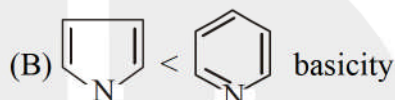
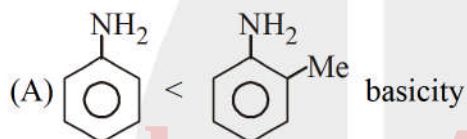
- (A) III and IV (B) II and III (C) I and II (D) I and III

Q.7 Which of the above compound(s) have L-configuration ?

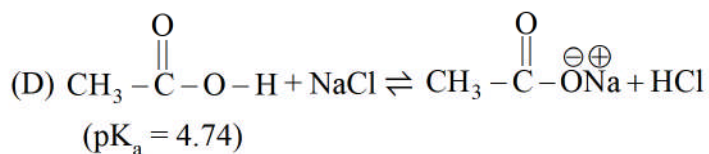
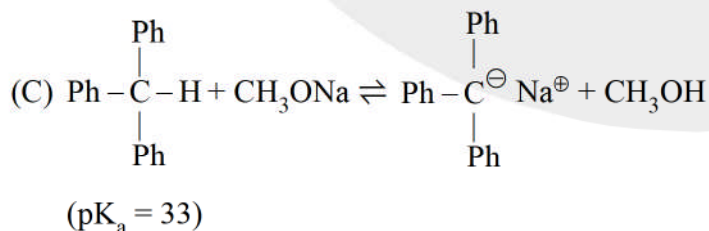
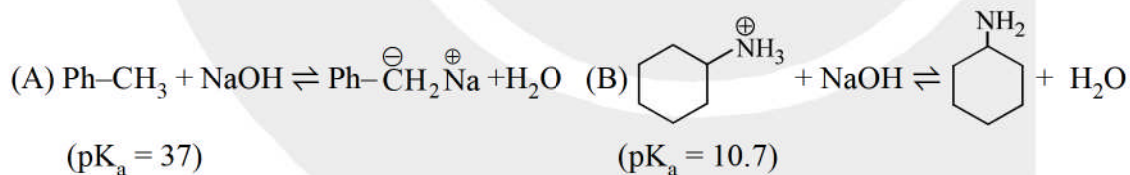
- (A) I and III (B) III only (C) III and IV (D) II and IV

More than one correct :

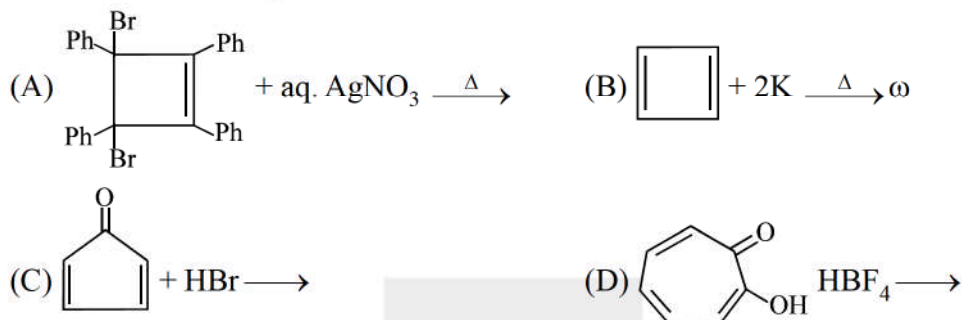
Q.8 Select correct options.



Q.9 In which of following reaction direction of equilibrium is in backward direction.

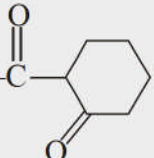


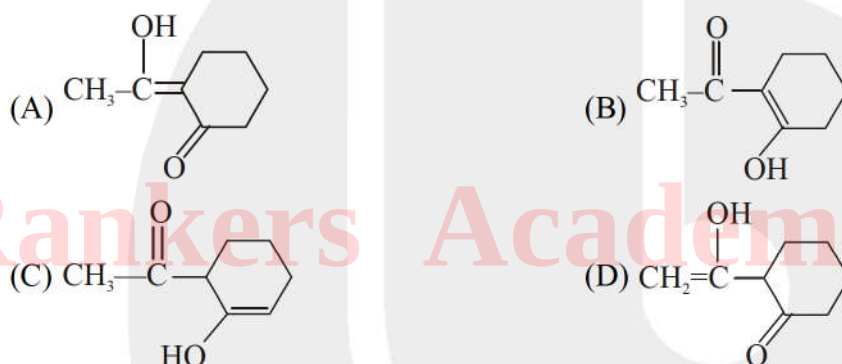
Q.10 Which of the following reaction will form stable ion.



Q.11 Select true statement :

- (A) Gauche form of 3-hydroxy propanol is more stable than it's anti form
 (B) Anti form of n-butane (across $\text{C}_2 - \text{C}_3$) is more stable than it's Gauche form.
 (C) At room temperature mole fraction of eclipsed form is almost zero.
 (D) Dipole moment of 1,2-dichloro ethane decreases on heating.

Q.12 Tautomers of the compound  is / are

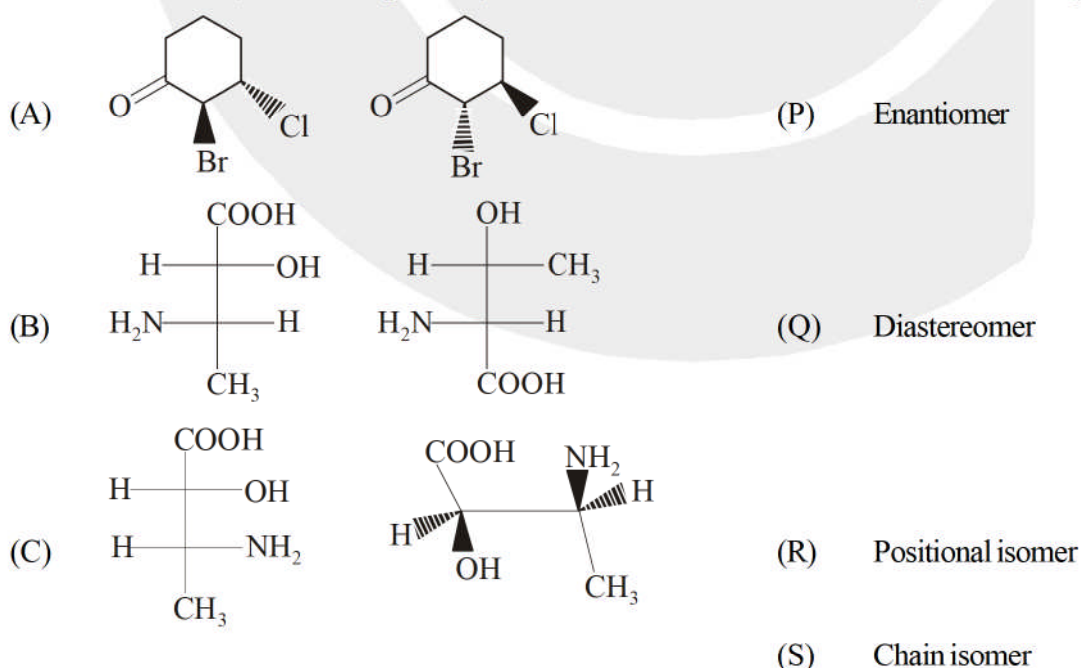


Match the column :

Q.13

Column I(Pair of compounds)

Column II(Relationship)



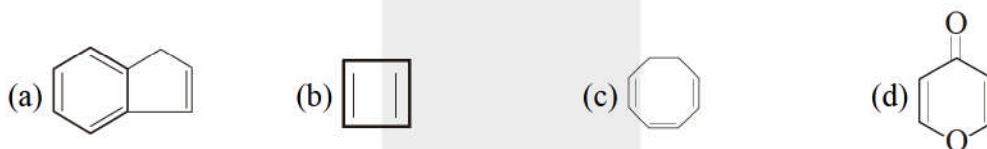
Subjective :

Q.14 If ΔH_H (Heat of hydrogenation) of pent-1-ene is 30 Kcal approx. & that of penta-1,3-diene is 56 Kcal approx. then what is the

- (a) Resonance energy of penta-1,3-diene. (x)
- (b) What is the ΔH_H per π bond of penta-1,3-diene (yz)
- (c) What is the D.B.E. (double bond equivalent) of penta-1,3-diene. (v)

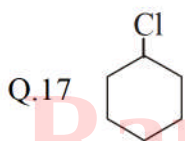
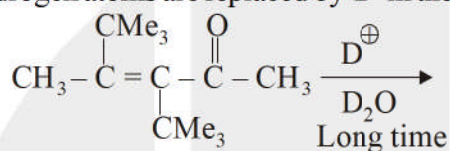
Represent your answer as xyzv.

Q.15 Identify aromatic, non aromatic and anti-aromatic :



Write (1) for Aromatic, (2) for Non-aromatic and (3) for Anti-aromatic compound in OMR sheet.

Q.16 How many Hydrogen atoms are replaced by D in the following tautomerization.



$\xrightarrow[\text{hv}]{\text{Cl}_2}$ number of theoretically possible dichloro derivatives

ANSWER KEY

Q.1	C	Q.2	C	Q.3	D	Q.4	C	Q.5	CD
Q.6	C	Q.7	C	Q.8	BCD	Q.9	ACD	Q.10	ABD
Q.11	ABC	Q.12	ABCD	Q.13	[(A) P (B) R (C) Q]				
Q.14	[4282]	Q.15	[1321]	Q.16	[6]	Q.17	[9]		

